README.md

Powder Physics

A real-time 2D powder physics simulation focused on granular materials, cellular interactions, and emergent behavior.

How to Use

1. **Download** the .jar release
2. **Extract** the downloaded folder
3. **Run** the executable
   - If you encounter a "Permission Denied" error, run the following command to set the correct permissions:
     `bash chmod 700 <path to file>`

Features

Grid-Based World

- The simulation runs on a 2D grid
- Each cell in the grid represents a single entity
- Designed for deterministic and scalable updates

Material Categories

Supports multiple material categories, each with unique behavior:

Solids

- Sand
- Stone
- Dirt

Liquids

- Water
- Oil
- Honey
- Lava
- Acid

Gases

- Smoke

Entity Composition System

- Each grid entity is built using composition
- Makes the system modular, scalable, and easy to extend
- New material types can be added without modifying existing logic

Physics & Movement

- Velocity-based movement for all entities
- Per-type collision checks and interaction rules
- Realistic material behavior determined on a type-by-type basis

User Controls

- Simple and intuitive menu for selecting material types
- Adjustable brush size for precise or large-scale placement
- Real-time interaction with the simulation

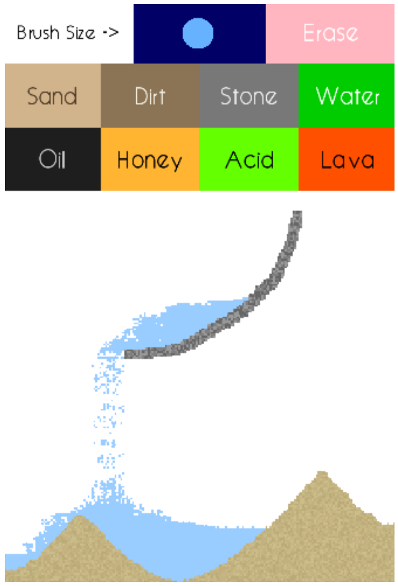
Multithreaded Simulation

- The grid is split into column-based chunks
- Multiple columns are processed simultaneously
- Improves performance and scalability on larger worlds

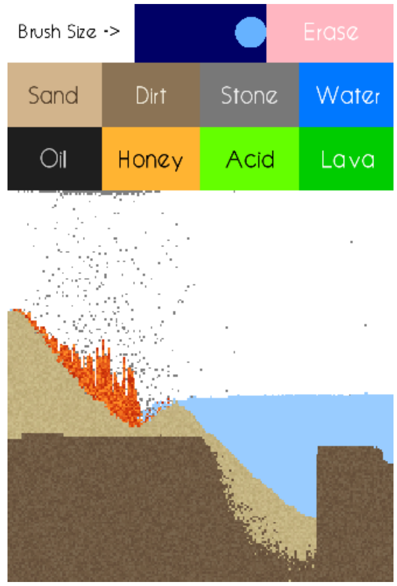
Pictures

Explore the simulation with realistic physics interactions.

Example 1



Example 2



Example 3